

Anaerobic Speed Reserve: A Key Component of Elite Male 800-m Running

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Purpose: In recent years (2011–2016), men’s 800-m championship running performances have required greater speed than previous eras (2000–2009). The “anaerobic speed reserve” (ASR) may be a key differentiator of this performance, but profiles of elite 800-m runners and their relationship to performance time have yet to be determined. **Methods:** The ASR—determined as the difference between maximal sprint speed (MSS) and predicted maximal aerobic speed (MAS)—of 19 elite 800- and 1500-m runners was assessed using 50-m sprint and 1500-m race performance times. Profiles of 3 athlete subgroups were examined using cluster analysis and the speed reserve ratio (SRR), defined as MSS/MAS . **Results:** For the same MAS, MSS and ASR showed very large negative (both $r = -.74 \pm .30$, $\pm 90\%$ confidence limits; very likely) relationships with 800-m performance time. In contrast, for the same MSS, ASR and MAS had small negative relationships (both $r = -.16 \pm .54$; possibly) with 800-m performance. ASR, 800-m personal best, and SRR best defined the 3 subgroups along a continuum of 800-m runners, with SRR values as follows: 400–800 m ≥ 1.58 , 800 m ≤ 1.57 to ≥ 1.48 , and 800–1500 m ≤ 1.47 to ≥ 1.36 . **Conclusion:** MSS had the strongest relationship with 800-m performance, whereby for the same MSS, MAS and ASR showed only small relationships to differences in 800-m time. Furthermore, the findings support the coaching observation of three 800-m subgroups, with the SRR potentially representing a useful and practical tool for identifying an athlete’s 800-m profile. Future investigations should consider the SRR framework and its application for individualized training approaches in this event.

Keywords: middle distance running, training science, maximal sprint speed, maximal aerobic speed

Preparation for 800-m running represents a unique challenge to the middle-distance coach. With close interplay required between aerobic and anaerobic/neuromuscular physiology, athletes with distinctly different profiles have an opportunity for success in the event. Recently, a “changing of the guard” in the men’s championship 800-m event was revealed, whereby from 2011 onwards World and Olympic medalists were shown to run predominantly a “gun-to-tape” type race tactic in the final,¹ requiring 100-m sectors that are 0.5 m/s faster than in 2000–2009. This raises important questions pertaining to the mechanical speed range required in top athletes relative to their aerobic capabilities.

Previous studies of national and international caliber 800- and 1500-m runners reveal opposing findings regarding the physiological requirements of 800-m running. For example, Ingham et al² reported that $\dot{V}O_{2\max}$ and running economy explained 95.9% of running performance in 800- and 1500-m runners; however, no speed and power measures were collected. In contrast, Bachero-Mena et al³ showed very large relationships between 800-m performance and sprints over 20 m ($r = .72$) and 200 m ($r = .84$), yet aerobic markers were not reported. Several reasons may explain these contrasting outcomes. First, athletes with diverse physiological profiles may be competing in the same event. For example, Schumacher and Muller⁴ showed in Olympic gold medal-winning team pursuit cyclists that the first and second position riders presented with markedly different anaerobic and aerobic

physiological profiles, yet produced similar individual pursuit performance times (4:18 vs 4:19). Accordingly, it is possible that more aerobic-based 800-m runners demonstrate stronger relationships between aerobic-measured variables and performance, whereas more sprint-based 800-m athletes present stronger correlations with anaerobic and neuromuscular qualities, depending on the random phenotype predominance of the sample. Second, it is possible that a cultural endurance-focused training approach bias has contributed to a production of studies from the more endurance-based 800-m running subgroup. Third, although the 800 and 1500 m have historically been considered as “similar events,”² this belief may require reassessment in light of recent tactical evolution.¹ Indeed, heterogeneity of performance standard within “elite” samples may be misleading when it comes to differentiating elite athlete makeup. For example, conclusions are often drawn on elite performance when samples may contain only 1 or 2 truly elite runners (800-m performance $\leq 1:46$).^{3,5}

Middle-distance coaches have long spoken of 3 subgroups of middle-distance runners. These include: (1) speed types (400- to 800-m specialists), (2) 800-m specialists, and (3) endurance types (800- to 1500-m specialists).^{6,7} Understanding the athlete subgroup has substantial influence on the coach’s plan, training program, and coaching approach. In contrast, the sport science literature has traditionally treated the 800-m cohort as a single athlete type, without assessing individual characteristics that might form a coach’s subgroup. Although a minimum level of both aerobic and neuromuscular qualities would be required for success in any elite 800-m runner, a deficiency in either component is likely balanced by a strength in the other to achieve a 2-lap performance.⁸ However, without information defining this variability, clarity of training methods for this event group cannot be established to the degree that they have been with runners training for the 1500- to 10,000-m events.⁹

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Sanders et al¹⁰ recently showed the usefulness of the anaerobic power reserve construct for predicting sustainable power performance across 4 professional road cyclists with largely diverse peak power profiles (range = 1036–1525 W). Therefore, the anaerobic speed reserve (ASR), defined as the speed range an athlete possesses between velocity at $\dot{V}O_2\text{max}$ ($v\dot{V}O_2\text{max}$) in the laboratory (or maximal aerobic speed [MAS] in the field¹¹) and maximal sprint speed (MSS),¹² may likewise prove a useful tool to better understand the apparent diversity of mechanical profiles across the 800-m event group. In addition, ASR may provide the coach and sport scientist a profile for assessing an athlete's mechanical limits supported by their metabolic systems (aerobic and anaerobic) as well as for tracking progress in training.¹³

Therefore, the aims of the present study were to (1) determine MSS and ASR and their relationship with 800-m performance in an elite middle-distance cohort and (2) using the ASR construct, investigate the athlete profiles within the event and offer possible solutions for coaches and scientists to be able to better categorize 800-m athletes.

Methods

Study Overview

To perform this study, the primary researcher traveled to locations around the world to test participants at their local training venue during the late precompetition/competition phase of the 2017 athletics season. At each training location, athletes were tested for their MSS. Within 6 weeks of the MSS assessment, an outdoor 1500-m race used to estimate MAS, was performed in competition.

Participants

A total of 19 elite 800- and 1500-m specialists representing 5 different continents (Africa, Europe, North America, Oceania, and South America) participated in this study (Table 1). The study inclusion criteria included an 800-m personal best (PB) of $\leq 1:47.50$, and/or a 1500-m PB of $\leq 3:40$, as guided by USA track and field World Championship trial standards (2017). Season's bests (SB) were used throughout the analysis to better reflect an athlete's current shape (eg, PB could be up to 3 years prior). Each athlete provided written informed consent to participate in the study, which was approved by the Auckland University of Technology Ethics Committee.

Performance Testing

Maximal Aerobic Speed. For MAS assessment, a “gun-to-tape” 1500-m race performed within 6 weeks of MSS assessment was reflective of an athlete's absolute time-trial capacity and current aerobic fitness. In line with the periodization phase described, data collection aligned well with the period where athletes were targeting qualifying times and “gun-to-tape” style races led by a pacemaker to 1000 to 1200 m; an aspect that would also have likely enhanced the reliability of 1500-m times. From these times, MAS was predicted using the 1500-m race performance equation of Bellenger et al¹¹:

$$\text{MAS} = \text{TT}_s (0.766 + 0.117 [\text{TT}_d])$$

where TT_s was the athlete's average 1500-m speed (in kilometer per hour) and TT_d was 1.5 km.

Maximal Sprint Speed. Upon arrival at the track, participants were informed of the rationale for the 50-m MSS assessment and maximal nature of the testing protocol. Athlete performance was measured using a sports radar device (Stalker ATS II System; Radar Sales, Richardson, TX) over a 50-m sector on the track straight. The device was placed in the middle of 2 lanes, 2 m behind the start line, and on a tripod resting 1.5 m from the ground. For live capture of the athlete's acceleration trace, the radar was operated remotely from a laptop to remove the possibility of manual use variability, using a method that has been shown to be highly reliable ($\text{CV} = 1.1\%$) against gold-standard force plates.¹⁴ Instantaneous radar was used to extract MSS (in meter per second) and split time (in seconds) from each effort, sampling at 46.9 Hz. Custom-built software (Goldmine, HPSNZ, NZ), was used to remove postprocessing error of the acceleration trace from manual inspection of erroneous data points. Previous investigations have shown the reliability limitation of postprocessing with “LabVIEW” (Build version: 11.0, National Instruments Corp, Austin, TX) software.¹⁵

Owing to the experienced status of the athletes, and cultural differences in warm-up, a semistructured framework was used to provide consistency across sites. Here, instructions were to take 10 to 15 minutes to prepare for a maximal effort, incorporating individual needs to feel ready to go. Athletes were familiarized with the standing start position and instructed to place 1 foot in front of the other (athlete's preference), with no backward oscillation, though a forward lean into the movement was permitted into the first forward step.

Boundaries for the warm-up included some pulse-raising activity (jogging), some drills (A skips, B skips, etc), time for any other exercises athletes required and 2 to 3 progressive strides in flats, before transitioning into race spikes, where athletes were asked to rehearse the standing start in 1 to 2 maximal efforts to 30 m. Athletes performed the assessment in spikes ($n = 17$) or race flats ($n = 2$).

Once ready to go, an instruction of “on you” was provided for the athlete to accelerate in their own time maximally through to the end of the cones, along with the line at the center of the 2 lanes. Athletes performed 2 to 3 maximal efforts with ~ 3 minutes rest on rotation at the end of the athlete testing line. The primary variable of interest taken from the radar for analysis was the MSS, representing the ceiling of the athlete's ASR. MSS assessments were conducted where possible in an indoor location, but where not possible (6 sites), a wind gauge (Kestrel 5100; Nielsen-Kellerman Company, Boothwyn, PA) was used to measure wind speed. All MSS assessments outdoors were captured with 0.5 ± 0.3 m/s tailwind. Environmental conditions indoors ($25.2^\circ\text{C} \pm 3.5^\circ\text{C}$, $51.4 \pm 10.7\%$ RH) and outdoors ($26.2^\circ\text{C} \pm 5.0^\circ\text{C}$, $42.7 \pm 22.8\%$ RH) were similar. One site was at 580-m altitude with all others at sea level.

Speed Reserve Ratio

The speed reserve ratio (SRR) was developed from the ASR construct as a potentially practical tool for coaches to display individual athlete profiles in 1 variable, as used in team sports.¹⁶ Whereby:

$$\text{SRR} = \text{MSS (km/h)} / \text{MAS (km/h)}$$

Data Analysis

Data are presented as mean $\pm$ 90% confidence limits (CL) unless otherwise stated. The relationships between 800-m SB performance and MSS, MAS, and ASR ($n = 10$) were determined from partial correlations using SAS 9.4 (SAS Institute, Cary,

Table 1 Description of the Participants in the Study

Category of athlete	Regions represented	Highest competition representation	Other details	800-m PB, mm:ss.ms (mean [SD])	1500-m PB, mm:ss.ms (mean [SD])
International, n = 8	Europe, North America, South America, and Oceania	Olympic Games/World Championships, n = 6 World Indoor Championships, n = 1 World Relay Championships, n = 1	Includes 1 × world-record holder, 2 × national-record holders	1:45.55 (1.18)	3:46.69 (8.20)
European, n = 3	Europe	European U23 Cross Country European U23 Outdoor Championships European U20 Outdoor Championships	1 × medalist	1:47.07 (0.15)	3:39.93 (3.53)
National, n = 2	Oceania	National Championship	5 × national champion 1 × national medalist	1:47.80 (1.13)	3:42.34 (5.04)
Collegiate, n = 6	Africa, Europe, and North America	World University Games, n = 3 NCAA Outdoor Championship, n = 3	World University Games 2 × medalist Includes African championship finalist	1:47.90 (1.84)	3:41.40 (3.04)

Abbreviation: PB, personal best; NCAA, National Collegiate Athletic Association.

NC) and described by magnitude-based inferences.¹⁷ As alluded to in the introduction, the merging of elite and subelite athlete data into the same analysis^{2,3} may explain disparate outcomes reported previously. Therefore, two 1500-m specialists who did not have an 800-m SB on record were removed from this part of the analysis. Five athletes with a SB of >1:47.50 for 800 m were also removed with times outside cutoffs for elite status.

A *k*-means cluster analysis was performed using SAS 9.4 to investigate the variation in physiological and performance characteristics of world-class 800-m runners. Athletes excluded from partial-correlation analysis (due to the 800-m focus) were included for subgroup clustering (*n* = 19). Instruction was given to fit the collected variables into 3 clusters, as per the aforementioned coach observations. MAS, MSS, ASR, SRR, body mass, 800-m PB and SB, 1500-m PB and SB obtained through testing, questionnaire, or competition data collection were standardized and run through the cluster analysis to understand which best explained any differences between groups.

Variables were excluded based on their ability to explain variation between clusters, with the strongest relationships (highest R^2 value) used for final subgroup determination. Differences between clusters were determined using magnitude-based inferences. The following threshold values used for effect size statistics were ≥ 0.2 (small), ≥ 0.6 (moderate), ≥ 1.2 (large), and ≥ 2.0 (very large). The smallest worthwhile change for maximum velocity was determined as the SD of all 19 athletes' MSS, multiplied by the effect size.¹⁷ Moderate thresholds were applied across all variables to acknowledge variability in the MAS equation.¹¹

To explore the individual variation specifically in the 800 m, the SRR was used. For this analysis, 10 athletes with SB $\leq 1:47.50$, who also had a MAS marker within the 6-week window were assessed. 800-m SB times of these participants ranged from 1:44.50 to 1:47.36.

Results

ASR and 800-m Performance Relationships

Participant details are described in Table 1. MSS and ASR showed very large negative (both $r = -.74 \pm .30$; very likely) relationships with 800-m performance time for the same MAS. In contrast, ASR and MAS had small negative relationships (both $r = -.16 \pm .54$, possibly) with 800 m when MSS was constant (Figures 1A–1C).

800-m Subgroup Variation

The ASR, SRR, and 800-m PB accounted for the greatest variation between the 3 clusters of 800-m athletes ($R^2 = .87$; very large). Subgroup performance characteristics are shown in Table 2.

The MSS of 400- to 800-m athletes was faster than the 800-m specialists (1.8 ± 0.6 km/h, moderate, very likely), and 800- to 1500-m athletes (4.0 ± 0.4 km/h, very large, very likely). The 800-m specialists had faster MSS than 800- to 1500-m athletes (2.2 ± 1.5 km/h, large, likely). MAS in 400- to 800-m athletes was slower than both 800-m specialists (-0.5 ± 0.5 km/h, moderate, likely) and 800- to 1500-m athletes (-0.8 ± 0.4 , large, very likely). MAS of 800-m specialists was slower than 800- to 1500-m athletes (-0.3 ± 0.3 , moderate, possibly). ASR of 400- to 800-m athletes was larger than 800-m specialists (2.3 ± 0.7 km/h, large, very likely) and 800- to 1500-m athletes (4.3 ± 1.2 km/h, very large, most likely). ASR of 800 specialists was larger than 800- to 1500-m athletes (2.0 ± 1.2 km/h, moderate, likely).

The SRR had a large relationship with 800-m performance ($r = .53 \pm .43$, likely) whereby faster 800-m athletes had a higher ratio. In addition, body mass showed a large relationship with SRR ($r = .62 \pm .34$, very likely). 400- to 800-m athletes were heavier than 800-m specialists (6.4 ± 7.8 kg, moderate, possibly) and 800- to 1500-m athletes (5.8 ± 11.2 kg, moderate, possibly), with trivial differences between 800-m specialists and 800- to 1500-m athletes (-0.6 ± 11.1 kg, possibly).

Discussion

In the present study, we examined for the first time, the role of the ASR in elite male 800-m running performance, in an era where faster top speed appears to be a critical performance requirement.¹ Our findings confirm this observation, with a greater MSS (and therefore ASR) strongly correlated with a faster 800 m. Importantly, for the same MSS, having a greater MAS or ASR was not strongly related to changes in 800-m time. These results support the notion that at an elite level, faster 800-m runners have a larger ASR, which is related to a higher MSS (Figures 1A and 1C), along with an already established minimum level of estimated MAS. In addition, in agreement with longstanding coaching observations,^{6,7} we reveal the profiles of three 800-m athlete subgroups, described along a continuum herein as 400 to 800 m (speed types), 800 m (specialists), and 800 to 1500 m (endurance types; Table 1). Finally, we present the SRR construct (Figure 2) as a practical and easily implemented tool to support a coach's identification of the 800-m athlete subgroup, which may aid training approaches and event specialization.

The unique nature of the global elite study sample (Table 1) represents a critical addition to the middle-distance literature, with high relevance to coaches, athletes, scientists, and sports federations. Importantly, we confirm the role of ASR (through the function of larger MSS) as a key performance indicator of elite male 800-m running. Indeed, for the same MSS, having a greater MAS or ASR was not strongly related to changes in 800-m time. The paradigm offered by our analysis should consider that once a certain aerobic standard is reached, MSS becomes a differentiating factor in elite 800-m runners. In agreement with Bachero-Mena et al,³ we found a very large relationship between MSS and 800-m running performance (Figure 1A). The small relationship with MAS contrasts the study by Ingham et al,² who studied athletes with slower performance times ($1:48.9 \pm 2.4$), where perhaps the aerobic component may be a more important differentiator of slower ($\geq 1:47.50$) performance times. As we have shown, in an elite 800-m running cohort, the strong relationships between 800-m performance times and MSS (Figure 1A) and ASR (Figure 1B) demonstrate the importance of possessing advanced speed characteristics alongside an already well-developed aerobic capability.

It appears that to be competitive in the modern-day elite 800 m era, an MSS of ~ 10 m/s is required to cope with the high-speed demands in the first 200 m of the race (Table 2).¹ Notwithstanding, the complex phenomenon of any middle-distance performance,¹⁸ amid tactics, trips, mistimed training, injury, illness, and other uncertainties that occur,^{19,20} our data suggest that at the elite level, a baseline level of MSS/MAS characteristics are required to handle surging in slow, fast, or moderate paced 800-m events. Critically, considering the energetic demands of the men's 800-m (66% aerobic),²¹ neither aerobic nor anaerobic/neuromuscular components of training can be neglected.

Historically, the most common coaching approaches for differentiating 800-m athletes into subgroups involves using 400-m

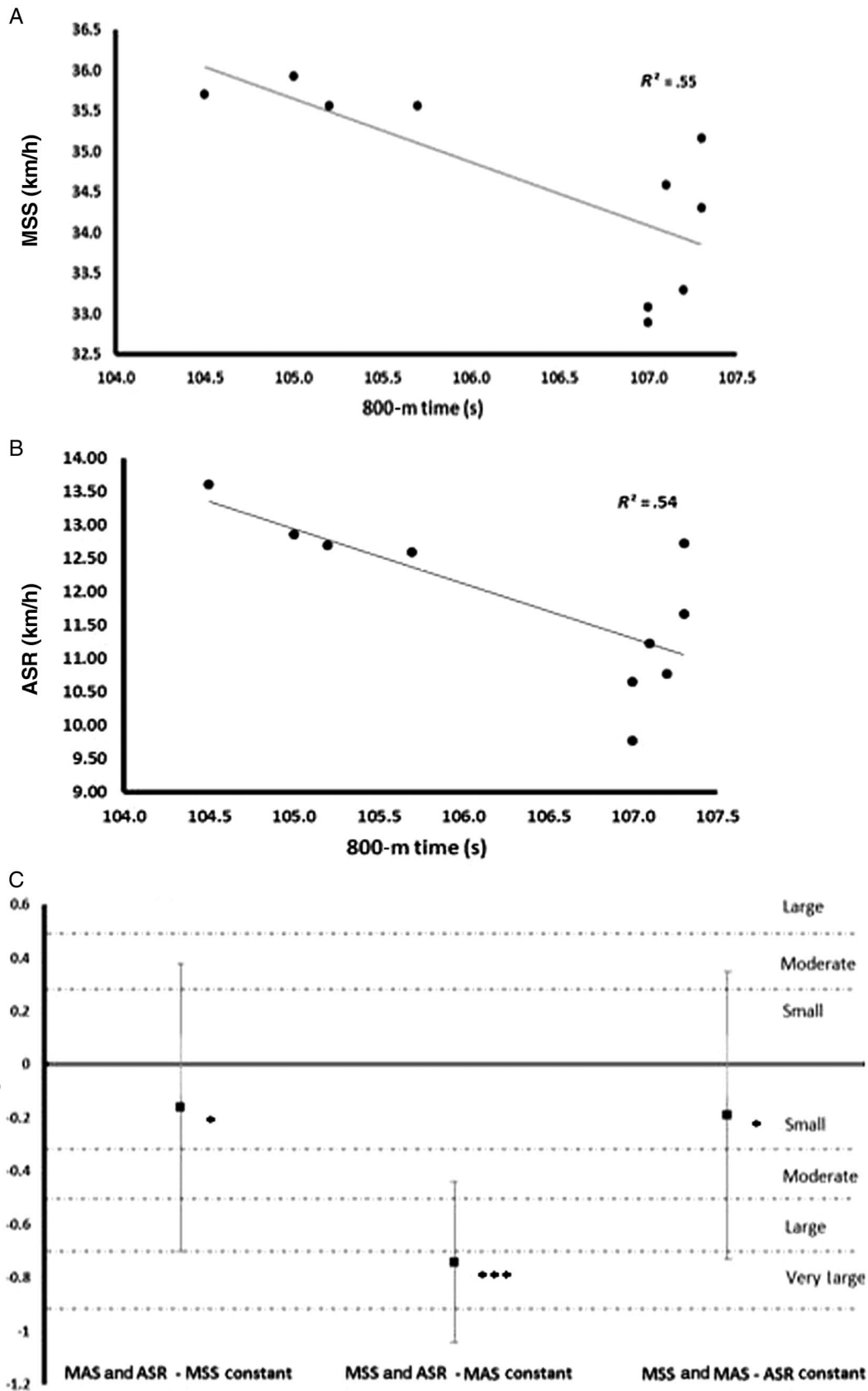


Figure 1 — Relationship between (A) MSS and (B) ASR with 800-m season's-best race performance in 10 elite-class male 800-m runners. (C) Partial correlation magnitudes with 90% confidence limits; gray area represents trivial relationship. Change *possibly substantial, **likely substantial, and ***very likely substantial. ASR indicates anaerobic speed reserve; MAS, maximal aerobic speed; MSS, maximal sprint speed.

Table 2 Performance (mm:ss.ms) and Profile Characteristics of the 800-m Subgroups (N = 19), Mean (SD)

	400- to 800-m athletes (n = 10)	800-m specialist (n = 6)	800- to 1500-m athletes (n = 3)
800-m personal best	1:46.21 (1.16)	1:46.37 (1.43)	1:49.53 (1.28)
1500-m personal best	3:44.05 (4.33)	3:42.13 (3.87)	3:38.89 (0.87)
Body mass, kg	72.2 (8.3)	65.8 (8.3) ^a	66.4 (6.9) ^{a,e}
MSS, km/h	35.48 (0.30) ^g	33.68 (0.63) ^c	31.49 (0.99) ^{c,f}
MAS, km/h	22.41 (0.62)	22.76 (0.50) ^b	23.21 (0.06) ^{c,e}
ASR, km/h	14.46 (1.00) ^g	12.12 (0.61) ^c	10.13 (0.76) ^{d,f}
SRR	≥1.58	≤1.57 to ≥1.48	≤1.47 to ≥1.36

Abbreviations: ASR, anaerobic speed reserve; MAS, maximal aerobic speed; MSS, maximal sprint speed; SRR, speed reserve ratio.
^a Possibly substantial difference from 400–800 m. ^b Likely substantial difference from 400–800 m. ^c Very likely substantial difference from 400–800 m. ^d Most likely substantial difference from 400–800 m. ^e Possibly substantial difference from 800-m specialist. ^f Likely substantial difference from 800-m specialist. ^g Very likely substantial difference from 800-m specialist.

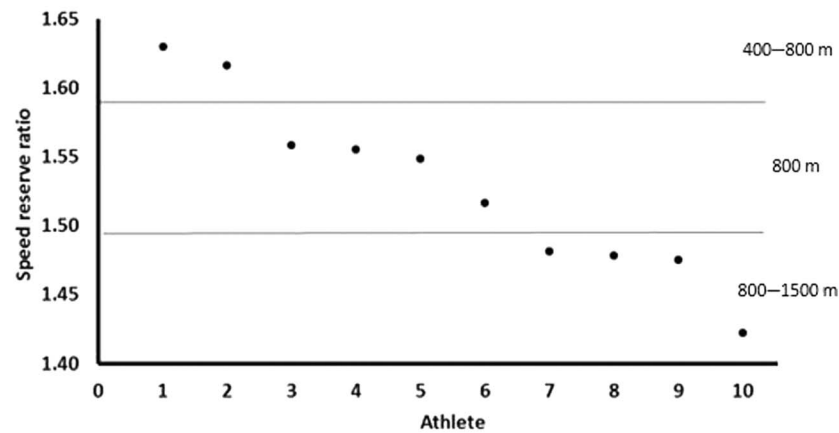


Figure 2 — Speed reserve ratio (maximum sprint speed [km/h]/maximal aerobic speed [km/h]) of 10 elite male 800-m runners. Lines depict 800-m subgroups from cluster analysis.

and 1500-m PB times to segregate an athlete into 400- to 800-m or 800- to 1500-m subgroups.^{6,7} The present investigation offers the SRR as an additional tool for classifying athletes into subgroups (Table 2; Figure 2). MAS differences between 400 to 800 m and 800-m specialists, and 800-m and 800- to 1500-m subgroups were moderate, whereas they were large between 400- to 800-m and 800- to 1500-m athletes. However, the differences in MSS were much greater between subgroups (Table 2). Therefore, SRR may be the most effective metric for easily identifying an athlete’s subgroup, as evidenced by the cluster analysis, which revealed that ASR, SRR, and 800-m PB accounted for the greatest amount of subgroup variation. Further studies investigating middle distance running should consider stating the distribution of athlete subgroups in the methodology sections and perform data analysis per subgroup to provide readership with outcomes of interventions (eg, training or nutritional) on specific subgroups with similar profiles (Table 2).

Many questions remain across 800-m subgroup characterization, in terms of how mechanical and metabolic components may explain these results. The critical speed describes the divide between steady state and nonsteady state exercise, with the finite work capacity above critical speed termed *D* prime (*D*′).²² *D*′ is our current best estimate of an athlete’s so-called anaerobic capacity, a measurement that has challenged physiologists for years.²³ Previously,²⁴ it was shown that Finnish national 800- and 1500-m distance runners and US 400-m athletes (PB range: 44–52.5 s) had superior anaerobic work capacity (as defined from the maximal

anaerobic running test) compared with long-distance runners and control (sprinters and jumpers) groups. The 400-m athletes had superior anaerobic work capacity and the highest MSS in comparison to the national Finnish 800- and 1500-m athletes; however, individual event comparisons were not provided. A fast MSS determines the proportion of ASR an athlete can work at and may relate to high-intensity training tolerance.²⁵

Body mass showed a large positive relationship with SRR (*r* = .62), which may be explained by the underlying ground force characteristics, with MSS ultimately limited by the impulse an athlete can produce.²⁶ Van Der Sward²⁷ showed that fast-twitch muscle fiber composition and vastus lateralis muscle volume explained 65% of the normalized peak power output in cycling. Therefore, greater muscle mass differences (inferred from body mass measurement) between subgroups may explain part of their different MSS capability (Table 2). Furthermore, muscle composition differences between the subgroups have implications for VO₂ kinetics, with slower oxygen flux through type IIa and IIx fibers,²⁸ as well as smaller capillary density and electron transport chain enzymes versus type I fibers.²⁹ Differences in metabolic efficiency (lower efficiency in type II fibers) may have implications for metabolic perturbation during exercise intensities above critical speed,³⁰ such as in 800-m racing, therefore, reiterating the need to characterize *D*′ (alongside MSS) in 800-m subgroups. We postulate that perhaps 800-m specialist athletes are event experts, in part due to a predominance of IIa fiber types, which provide the unique blend of higher force generation characteristics than type I fibers,

and greater metabolic efficiency than type IIx, though this warrants confirmation using noninvasive muscle fiber-type estimation alongside SRR.³¹

Limitations

Our current methods used a 1500-m race for MAS prediction, potentially creating bias toward a better MAS prediction in 800 m rather than 1500-m specialists. With the current methods and logistics of the study collection, a race (that was already in most athlete's calendars), was deemed as the most practical method for capturing the MAS estimate in this elite sample during competition. However, specialization in competition today means that some 800-m athletes rarely, if ever, perform 1500-m races. The present sample also represents a distribution that unintentionally reflects clustering around the qualifying mark for nationals (1:47.50), similar to the Bannister 4-minute mile phenomenon,³² in addition to the difficulty of capturing World-Class participants for a research study. As such, this should be kept in mind with study interpretation.

The unique nature of this investigation into the speed characteristics of elite athletes in their various locations meant that laboratory assessment of $\dot{V}O_{2\max}$ was not practically possible, so a sound prediction method was a necessity. In this regard, it is also important to highlight that the variability of elite middle-distance racing is only ~1%,³³ far less than typical metabolic cart measurements (coefficient of variation $\dot{V}O_2$, ~4.7%³⁴). In addition, athletes may not always produce "true maximum" results in laboratory settings.¹⁷ Although unique, we believe our methodology provides a robust level of ecological validity and practicality for athletes and coaches.³⁵ Furthermore, the scientific literature has a comprehensive understanding of the aerobic determinants of elite middle-distance running, but data are scarce with respect to the MSS characteristics of elite 800-m runners.

Practical Application

An athlete's ASR and SRR showed the greatest variation between the 3 subgroups of elite male 800-m runners, and these variables, therefore, represent practical markers that coaches can easily measure to categorize their athletes' subgroup profiles. Importantly, many training studies show large individual variation in response to a given intervention, with the "responder and nonresponder concept" often attributed to the outcome.³⁶ Another explanation may be that for some athlete profiles, the stimulus provided could be inappropriate. For example, it is unlikely an anaerobic/neuromuscular-based athlete would respond to high densities of continuous aerobic work. The SRR framework (Table 2; Figure 2) could advance the profiling of athletes into subgroups based on their ASR characteristics, which may allow precise selection of more favorable training content. Such an approach has been successfully used in team sports,³⁷ and thus provides a fruitful opportunity for further understanding the individual training response required for different 800-m subgroups.

Conclusion

A larger ASR through the function of a faster MSS had the strongest relationship with elite 800-m performance. When MSS was held constant, MAS and ASR had only small relationships to differences

in 800-m time. In addition, the SRR, defined as the MSS/MAS, may represent a useful tool to identify an athlete's 800-m subgroup. Future investigations should consider the SRR framework and its application for individualized training approaches in this event.

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References

1. Sandford GN, Pearson S, Allen SV, et al. Tactical behaviors in men's 800-m Olympic and world-championship medalists: a changing of the guard. *Int J Sports Physiol Perform*. 2018;13(2):246–249. PubMed ID: 28488905 doi:10.1123/ijsp.2016-0780
2. Ingham SA, Whyte GP, Pedlar C, Bailey DM, Dunman N, Nevill AM. Determinants of 800-m and 1500-m running performance using allometric models. *Med Sci Sports Exerc*. 2008;40(2):345–350. PubMed ID: 18202566 doi:10.1249/mss.0b013e31815a83dc
3. Bachero-Mena B, Pareja-Blanco F, Rodríguez-Rosell D, Yáñez-García JM, Mora-Custodio R, González-Badillo JJ. Relationships between sprint, jumping and strength abilities, and 800 m performance in male athletes of national and international levels. *J Hum Kinet*. 2017;58(1):187–195. doi:10.1515/hukin-2017-0076
4. Schumacher YO, Mueller P. The 4000-m team pursuit cycling world record: theoretical and practical aspects. *Med Sci Sports Exerc*. 2002;34(6):1029–1036. PubMed ID: 12048333 doi:10.1097/00005768-200206000-00020
5. Ingham SA, Fudge BW, Pringle JS. Training distribution, physiological profile, and performance for a male international 1500-m runner. *Int J Sports Physiol Perform*. 2012;7(2):193–195. PubMed ID: 22634971 doi:10.1123/ijsp.7.2.193
6. Horwill FJ. Solving the 800 m puzzle. In: *Modern Athlete and Coach* (vol 343). Adelaide; 1996:48–51.
7. Gamboa J, Elrick R, Mora A, et al. NSA Round table—Speed in the 800 metres. *New Stud Athl*. 1996;11(4):7–22.
8. Daniels J. 800 meters. In: Martin B, Julie R, Carla Z. *Daniels Running Formula*. 2nd ed. Human Kinetics; 2005:201–212.
9. Tjelta LL. The training of international level distance runners. *Int J Sport Sci Coach*. 2016;11(1):122–134. doi:10.1177/1747954115624813
10. Sanders D, Heijboer M, Akubat I, Meijer K, Hesselink M. Predicting high-power performance in professional cyclists. *Int J Sports Physiol Perform*. 2017;12(3):410–413. PubMed ID: 27248365 doi:10.1123/ijsp.2016-0134
11. Bellenger CR, Fuller JT, Nelson MJ, Hartland M, Buckley JD, DeBenedictis TA. Predicting maximal aerobic speed through set distance time-trials. *Eur J Appl Physiol*. 2015;115(12):2593–2598. PubMed ID: 26242778 doi:10.1007/s00421-015-3233-6
12. Blondel N, Berthoin S, Billat V, Lensele G. Relationship between run times to exhaustion at 90, 100, 120, and 140% of $\dot{V}O_{2\max}$ and velocity expressed relatively to critical velocity and maximal velocity. *Int J Sports Med*. 2001;22(1):27–33. PubMed ID: 11258638 doi:10.1055/s-2001-11357
13. Weyand P, Lin J, Bundle M. Sprint performance-duration relationships are set by the fractional duration of external force application. *AJP Regul Integr Comp Physiol*. 2006;290:R758–R765. doi:10.1152/ajpregu.00562.2005

14. Samozino P, Rabita G, Dorel S, et al. A simple method for measuring power, force, velocity properties, and mechanical effectiveness in sprint running: simple method to compute sprint mechanics. *Scand J Med Sci Sports*. 2015;26:648–658. PubMed ID: [25996964](#) doi:[10.1111/sms.12490](#)
15. Simperingham KD, Cronin JB, Pearson SN, Ross A. Reliability of horizontal force-velocity-power profiling during short sprint-running accelerations using radar technology. *Sport Biomech*. 2017;3141: 1–12. doi:[10.1080/14763141.2017.1386707](#)
16. Mendez-Villanueva A, Buchheit M, Kuitunen S, Poon TK, Simpson B, Peltola E. Is the relationship between sprinting and maximal aerobic speeds in young soccer players affected by maturation? *Pediatr Exerc Sci*. 2010;22:497–510. PubMed ID: [21242600](#) doi:[10.1123/pes.22.4.497](#)
17. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41(1):3–13. PubMed ID: [19092709](#) doi:[10.1249/MSS.0b013e31818cb278](#)
18. Glazier PS. Towards a grand unified theory of sports performance. *Hum Mov Sci*. 2015;56:1–18. doi:[10.1016/j.humov.2015.08.001](#)
19. Raysmith BP, Drew MK. Performance success or failure is influenced by weeks lost to injury and illness in elite Australian track and field athletes: a 5-year prospective study. *J Sci Med Sport*. 2016; 19(10):778–783. PubMed ID: [26839047](#) doi:[10.1016/j.jsams.2015.12.515](#)
20. Jones A, Whipp B. Bioenergetic constraints on tactical decision making in middle distance running. *Br J Sports Med*. 2002;36: 102–104. PubMed ID: [11916890](#) doi:[10.1136/bjism.36.2.102](#)
21. Spencer MR, Gustin PB. Energy system contribution during 200- to 1500-m running in highly trained athletes. *Med Sci Sports Exerc*. 2001;33(1):157–162. PubMed ID: [11194103](#) doi:[10.1097/00005768-200101000-00024](#)
22. Vanhatalo A, Jones AM, Burnley M. Application of critical power in sport: what is the critical power concept? *Int J Sports Physiol Perform*. 2011;6:128–136. PubMed ID: [21487156](#) doi:[10.1123/ijspp.6.1.128](#)
23. Davison RR, van Someren KA, Jones AM. Physiological monitoring of the Olympic athlete. *J Sports Sci*. 2009;27(13):1433–1442. PubMed ID: [19813137](#) doi:[10.1080/02640410903045337](#)
24. Nummela A, Mero A, Stray-Gundersen J, Rusko H. Important determinants of anaerobic running performance in male athletes and non-athletes. *Int J Sports Med*. 1996;17(suppl 2):S91–S96. doi:[10.1055/s-2007-972907](#)
25. Buchheit M, Hader K, Mendez-Villanueva A. Tolerance to high-intensity intermittent running exercise: do oxygen uptake kinetics really matter? *Front Exerc Physiol*. 2012;3:406. doi:[10.3389/fphys.2012.00406](#)
26. Weyand PG, Davis J. Running performance has a structural basis. *J Exp Biol*. 2005;208(14):2625–2631. doi:[10.1242/jeb.01609](#)
27. van der Zwaard S, van der Laarse WJ, Weide G, et al. Critical determinants of combined sprint and endurance performance: an integrative analysis from muscle fiber to the human body. *FASEB J*. 2018;32(4):2110–2123. doi:[10.1096/fj.201700827R](#)
28. Crow MT, Kushmerick MJ. Chemical energetics of slow- and fast-twitch muscles of the mouse. *J Gen Physiol*. 1982;79(1):147–166. PubMed ID: [7061985](#) doi:[10.1085/jgp.79.1.147](#)
29. Jackman MR, Willis WT. Characteristics of mitochondria isolated from type I and type IIb skeletal muscle characteristics of mitochondria isolated from type I and type IIb skeletal muscle. *Am J Physiol*. 1996;270(39):C673–C678. doi:[10.1152/ajpcell.1996.270.2.C673](#)
30. Pringle JS, Doust JH, Carter H, et al. Oxygen uptake kinetics during moderate, heavy and severe intensity “submaximal” exercise in humans: the influence of muscle fibre type and capillarisation. *Eur J Appl Physiol*. 2003;89(3-4):289–300. PubMed ID: [12736837](#) doi:[10.1007/s00421-003-0799-1](#)
31. Baguet A, Everaert I, Hespel P, Petrovic M, Achten E, Derave W. A new method for non-invasive estimation of human muscle fiber type composition. *PLoS ONE*. 2011;6(7):e21956. doi:[10.1371/journal.pone.0021956](#)
32. Denison J. Inhibiting progress: the record of the four-minute mile. *Sport Hist*. 2006;26(2):280–288. doi:[10.1080/17460260600786930](#)
33. Hopkins WG. Competitive performance of elite track and field athletes. Variability and smallest worthwhile enhancements. *Sportscience*. 2005;9:17–20.
34. Crouter SE, Antczak A, Hudak JR, DellaValle DM, Haas JD. Accuracy and reliability of the ParvoMedics TrueOne 2400 and MedGraphics VO2000 metabolic systems. *Eur J Appl Physiol*. 2006;98(2):139–151. PubMed ID: [16896734](#) doi:[10.1007/s00421-006-0255-0](#)
35. Galbraith A, Hopker J, Lelliott S, Diddams L, Passfield L. A single-visit field test of critical speed. *Int J Sports Physiol Perform*. 2014;9(6): 931–935. PubMed ID: [24622815](#) doi:[10.1123/ijspp.2013-0507](#)
36. Mann TN, Lamberts RP, Lambert MI. High responders and low responders: factors associated with individual variation in response to standardized training. *Sport Med*. 2014;44(8):1113–1124. doi:[10.1007/s40279-014-0197-3](#)
37. Buchheit M. The 30–15 intermittent fitness test: 10 year review. *Myorobie J*. 2010;1(Top 14):1–9. [http://martin-buchheit.net/Dossiers/Buchheit-30-15IFT-10_yrs_review_\(2000-2010\).pdf](http://martin-buchheit.net/Dossiers/Buchheit-30-15IFT-10_yrs_review_(2000-2010).pdf).